

Solution Requirements

Date: 1 July 2026

Project Name: Customer Persona Generator - AI Powered Marketing Persona Application

Team ID: XXXXXX

Maximum Marks: 4 Marks

1. Functional Requirements (FR) - Part 1

The following table outlines the core functional requirements for the AI-powered persona application.

S.No	Requirement	Detailed Description	Priority
1	Authentication	Secure user authentication via OAuth (Google/LinkedIn) and multi-factor authentication (MFA) to ensure identity verification.	High
2	Authorization Levels	Implementation of Role-Based Access Control (RBAC) defining access permissions for Admins, Marketing Editors, and Guest users.	High
3	External Interfaces	Seamless integration with OpenAI GPT APIs for intelligent text generation and secure connectivity with external CRM databases.	Medium
4	Transactions Processing	Real-time handling of user-submitted marketing data, processing API tokens, and maintaining audit logs for all persona generation sessions.	High

1. Functional Requirements (FR) - Part 2

S.No	Requirement	Detailed Description	Priority
5	Reporting	System generates comprehensive PDF/ CSV exports of marketing personas, including visualization dashboards of key demographic trends.	Medium
6	Business Rules	The application utilizes proprietary logic constraints to map raw input data (e.g., surveys, social media data) into specific marketing persona segments.	High
7	Compliance	Strict adherence to global data protection regulations including GDPR, CCPA, and industry-specific privacy standards for customer data.	High
8	Other	User-friendly UI/UX, personalized dashboard configuration, and historical persona management features.	Low

2. Non-Functional Requirements (NFR)

S.No	Category	Description	Target Metric / Acceptance Criteria
1	Performance	The system must maintain low latency during AI processing tasks.	Average page response < 2 seconds; AI generation < 5 seconds.
2	Scalability	System design must support seamless horizontal scaling to accommodate growing user bases.	Supports 10,000+ concurrent active users.
3	Security	End-to-end data protection using industry-standard protocols.	AES-256 encryption at rest and TLS 1.3 in transit.
4	Reliability	High-availability infrastructure to ensure minimal downtime.	99.9% uptime per month with automated failover.
5	Usability	Accessibility-focused design that ensures ease of use.	Compliant with WCAG 2.1 Level AA standards.
6	Maintainability	Codebase must be modular and well-documented.	Automated unit testing with > 90% code coverage.